

Phase 2 — Cointegration Building Blocks

Estimators, theorems, and the three results that contradicted us

Project *regime-aware stat-arb & honest selection*

S&P 500 Utilities — 2014-2026

Mathematics stated and proved where the proof is short

1. What this document covers

Phase 1 produced a universe we can trust. Phase 2 produces the statistical machinery: the cointegration test, the hedge ratio, the half-life and the z-score. Four small functions, and four ways to be subtly and invisibly wrong.

This document states each estimator precisely, proves what can be proved in a few lines, and reports what was measured when theory alone was not enough. It opens with a correction to Phase 1, because that correction changes the numbers Phase 3 will be built on.

Three results that were not anticipated

The Phase 1 p-values were wrong. Running the ADF test on an OLS residual uses the wrong critical values and rejects 14.6 % of the time under the null instead of 5 %. Fixing it takes the 2016-2018 screen from 112 significant pairs to 47.

The half-life estimator does not return NaN on a random walk. It returns a finite, large value, and Kendall's bias formula predicts it: the phantom half-life grows linearly in the sample size, $h \approx 0.173 T$.

A rolling z-score does not damp the signal, it inflates it. The numerator shrinks but the denominator shrinks faster, and the strategy fires roughly 2.5 times too many entries at a 30-day half-life.

2. Setting and notation

Fix two securities A and B. Write $a_t = \log P_{A,t}$ and $b_t = \log P_{B,t}$ for their total-return log-prices, both assumed $I(1)$ in the sense of Phase 1. The *spread* with hedge ratio β and intercept α is

$$s_t = a_t - \beta b_t - \alpha.$$

The pair is cointegrated when some (β, α) makes (s_t) $I(0)$. Everything in Phase 2 is an answer to one of four questions: how to estimate β , how to test that s_t is $I(0)$, how fast s_t reverts, and how to turn s_t into a bounded trading signal without using the future.

Sample moments over an in-sample window of T observations are written S_{aa} , S_{bb} , S_{ab} for the centred sums of squares and cross-products. All estimation is in-sample; the walk-forward discipline that keeps it that way is Phase 3's responsibility, not this module's.

3. The hedge ratio

3.1 Ordinary least squares and its asymmetry

The OLS estimator minimises the sum of squared **vertical** deviations, which gives the familiar normal equations and

$$\hat{\beta}_{A|B} = \frac{S_{ab}}{S_{bb}}, \quad \hat{\alpha} = \bar{a} - \hat{\beta}_{A|B} \bar{b}.$$

The asymmetry this induces is not a subtlety to be mentioned in passing; it is exact and quantifiable.

Theorem 1 (OLS asymmetry)

For any sample with $S_{aa}, S_{bb} > 0$,

$$\hat{\beta}_{A|B} \cdot \hat{\beta}_{B|A} = R^2,$$

where R^2 is the coefficient of determination of either regression. In particular the two directions agree, in the sense that $\hat{\beta}_{A|B} = 1/\hat{\beta}_{B|A}$, if and only if $R^2 = 1$.

Proof.

Directly from the definitions,

$$\hat{\beta}_{A|B} \hat{\beta}_{B|A} = \frac{S_{ab}}{S_{bb}} \cdot \frac{S_{ab}}{S_{aa}} = \frac{S_{ab}^2}{S_{aa}S_{bb}} = r^2 = R^2,$$

r being the sample correlation. The second claim follows because $\beta_{A|B}^* = 1/\beta_{B|A}^*$ is equivalent to the product being 1. ■

The practical consequence is larger than intuition suggests. Measured on simulated pairs: at $R^2 = 0.93$ the two directions already differ by 7 % (1.416 against 1.518); at $R^2 = 0.11$ they differ by a factor of nearly ten (1.23 against 11.72). Same data, same pair, and a hedge ratio you could size a position on either way.

3.2 Total least squares is symmetric

Total least squares minimises **perpendicular** distance instead. For centred data and a unit direction vector u , the objective is

$$\min_{\|u\|=1} \sum_i \|x_i - (x_i' u) u\|^2 = \sum_i \|x_i\|^2 - \max_{\|u\|=1} u' S u,$$

with S the 2×2 scatter matrix. The minimiser is therefore the leading eigenvector of S , and the fitted slope is the ratio of its components.

Theorem 2 (TLS is exactly symmetric)

Write $D = S_{aa} - S_{bb}$ and $R = (D^2 + 4S_{ab}^2)^{1/2}$. The total-least-squares slopes are

$$\hat{\beta}_{A|B}^{\text{TLS}} = \frac{D+R}{2S_{ab}}, \quad \hat{\beta}_{B|A}^{\text{TLS}} = \frac{-D+R}{2S_{ab}},$$

and they satisfy $\hat{\beta}_{A|B}^{\text{TLS}} \cdot \hat{\beta}_{B|A}^{\text{TLS}} = 1$ identically, so the estimator does not depend on which leg is called dependent.

Proof.

The leading eigenvalue of S is $\lambda = ((S_{aa} + S_{bb}) + R)/2$, with eigenvector proportional to $(S_{ab}, \lambda - S_{bb})'$; the slope is the ratio of the second component to the first, which gives the stated formula. Exchanging the roles of A and B replaces D by $-D$ and leaves R unchanged. Hence

$$\hat{\beta}_{A|B}^{\text{TLS}} \hat{\beta}_{B|A}^{\text{TLS}} = \frac{(D+R)(-D+R)}{4S_{ab}^2} = \frac{R^2 - D^2}{4S_{ab}^2} = \frac{4S_{ab}^2}{4S_{ab}^2} = 1.$$

■

3.3 The decision, and the option that is forbidden

Option	Verdict
Fixed convention (alphabetical, a dependent)	Chosen for testing. Arbitrary but frozen, therefore not a researcher degree of freedom.
Test both directions, keep the better p-value	Forbidden. Doubles the number of tests and biases selection toward whichever direction happened to look better -- a multiple-testing problem disguised as a modelling choice.
Total least squares	Symmetric by Theorem 2, which is the honest choice when neither leg has privileged status. But the Engle-Granger p-value is built on an OLS first stage; substituting TLS would invalidate the critical values of Section 5. Kept as a robustness check.

4. Mean reversion: AR(1), Ornstein-Uhlenbeck, half-life

4.1 The discrete-time picture

Model the spread as a first-order autoregression around a level μ ,

$$s_t - \mu = \varphi (s_{t-1} - \mu) + \varepsilon_t, \quad \varepsilon_t \text{ i.i.d. } (0, \sigma_\varepsilon^2).$$

The process is stationary if and only if $|\varphi| < 1$, in which case its unconditional variance is $\sigma_\varepsilon^2 / (1 - \varphi^2)$. Mean reversion is the statement $\varphi < 1$; the value of φ measures its speed.

Proposition 1 (Geometric decay of a shock)

For every $k \geq 0$,

$$\mathbb{E}[s_{t+k} | s_t] = \mu + \varphi^k (s_t - \mu).$$

Proof.

By induction. The case $k = 0$ is trivial. Assuming the statement for k and using the tower property together with the recursion,

$$\mathbb{E}[s_{t+k+1} | s_t] = \mu + \varphi (\mathbb{E}[s_{t+k} | s_t] - \mu) = \mu + \varphi^{k+1} (s_t - \mu),$$

since the innovation is centred and independent of the past. ■

Definition 1 (Half-life)

The *half-life* h is the horizon at which the expected deviation from the mean has halved: the solution of $\varphi^h = 1/2$.

Proposition 2 (Half-life formula, and how to estimate it)

For $0 < \varphi < 1$,

$$h = -\frac{\ln 2}{\ln \varphi}.$$

In practice one regresses the change on the level,

$$\Delta s_t = a + b s_{t-1} + \varepsilon_t \iff s_t = a + (1 + b)s_{t-1} + \varepsilon_t,$$

so that $\varphi = 1 + b$, with $\mu = -a/b$.

Proof.

Take logarithms in $\varphi^h = 1/2$: $h \ln \varphi = -\ln 2$, and $\ln \varphi < 0$ for φ in $(0, 1)$. The regression identity is obtained by adding s_{t-1} to both sides of the first equation; setting the expected change to zero gives $a + b\mu = 0$. ■

4.2 The continuous-time picture, and why it is the same object

The Ornstein-Uhlenbeck process is the continuous-time analogue,

$$ds_t = \theta (\mu - s_t) dt + \sigma dW_t, \quad \theta > 0.$$

Theorem 3 (Solution and moments of the OU process)

The unique strong solution is

$$s_t = \mu + (s_0 - \mu)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-u)} dW_u,$$

with conditional moments

$$\mathbb{E}[s_t | s_0] = \mu + (s_0 - \mu)e^{-\theta t}, \quad \text{Var}(s_t | s_0) = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}),$$

and stationary law $N(\mu, \sigma^2/2\theta)$.

Proof.

Apply Ito's formula to $f(t, s) = e^{\theta t}(s - \mu)$:

$$df = \theta e^{\theta t}(s_t - \mu) dt + e^{\theta t} ds_t = \theta e^{\theta t}(s_t - \mu) dt + e^{\theta t}[\theta(\mu - s_t)dt + \sigma dW_t] = \sigma e^{\theta t} dW_t,$$

the drift terms cancelling exactly. Integrating from 0 to t and multiplying by $e^{-\theta t}$ gives the stated solution. The conditional mean follows because the stochastic integral is a centred martingale; the conditional variance follows from the Ito isometry,

$$\text{Var} = \sigma^2 \int_0^t e^{-2\theta(t-u)} du = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t}),$$

and letting t tend to infinity gives the stationary law. ■

Proposition 3 (Exact discretisation)

Sampling the OU process at a fixed step Δ yields exactly an AR(1) with

$$\varphi = e^{-\theta\Delta}, \quad \text{hence} \quad h = \frac{\ln 2}{\theta}.$$

The discrete and continuous half-lives are therefore the same number, not approximations of one another.

Proof.

Set $s_0 = s_t$ and $t = \Delta$ in Theorem 3. The conditional law of $s_{t+\Delta}$ given s_t is Gaussian with mean $\mu + (s_t - \mu)e^{-\theta\Delta}$ and a variance not depending on s_t , which is precisely an AR(1) recursion with $\varphi = e^{-\theta\Delta}$. Substituting into Proposition 2 gives $h = \ln 2 / \theta$. ■

5. Testing: Dickey-Fuller and Engle-Granger

5.1 The test, and why its distribution is not standard

Testing whether a series has a unit root means testing $\rho = 1$ in $X_t = \rho X_{t-1} + \varepsilon_t$, equivalently $b = 0$ in $\Delta X_t = a + b X_{t-1} + \varepsilon_t$. The augmented version adds lagged differences to absorb serial correlation in the innovations. The essential point is that the usual t-table is inapplicable.

Theorem 4 (Dickey-Fuller limiting distribution, 1979)

Under H_0 : $\rho = 1$ with i.i.d. centred innovations of finite variance,

$$T(\hat{\rho} - 1) \xrightarrow{d} \frac{\int_0^1 W dW}{\int_0^1 W^2 du} = \frac{\frac{1}{2}(W(1)^2 - 1)}{\int_0^1 W^2 du},$$

where W is a standard Brownian motion. The limit is **not** Gaussian, and the normalisation is T rather than the usual square root of T .

Proof.

Sketch. By Donsker's invariance principle the partial-sum process converges in the Skorokhod topology,

$$T^{-1/2}X_{[Tu]} \Rightarrow \sigma W(u).$$

Writing the OLS error as a ratio and rescaling numerator and denominator separately,

$$T^{-1} \sum_t X_{t-1} \varepsilon_t \Rightarrow \sigma^2 \int_0^1 W dW, \quad T^{-2} \sum_t X_{t-1}^2 \Rightarrow \sigma^2 \int_0^1 W^2 du,$$

and the continuous mapping theorem gives the ratio. The identity for the numerator is Ito's formula applied to W^2 . The faster T-rate is the usual super-consistency under a unit root: the regressor is of order square root of T rather than order one.

■

5.2 Engle-Granger: the residual is estimated, and that changes everything

The two-step procedure regresses a_t on b_t , then tests the residual for a unit root. The temptation is to feed that residual to a standard ADF routine. **That is the error made in Phase 1.**

Theorem 5 (Phillips and Ouliaris 1990 — stated without proof)

Under the null of no cointegration, the ADF statistic computed on the OLS residual $\hat{u}_t$ converges to a functional of a *demeaned and projected* Brownian motion whose law depends on the number of regressors. Its distribution is stochastically smaller (further into the left tail) than the Dickey-Fuller law of Theorem 4.

The intuition is worth more than the formal statement. OLS chose β to minimise the residual variance over an entire one-parameter family. The residual being tested is therefore the **most stationary-looking** linear combination available in that sample, not an arbitrary one. Testing it against critical values designed for an observed series necessarily over-rejects.

5.3 How large is the error? Measure it.

A theorem saying that critical values shift does not tell you whether the shift matters. Simulate under the null — two independent random walks, no cointegration by construction — and count rejections. A correct 5 % test rejects 5 % of the time.

```
2000 simulations under H0 (independent random walks, n = 750)

adfuller on the OLS residual : 14.60 % rejections    <- should be 5 %
coint (Engle-Granger values) :  4.15 % rejections

false-positive inflation factor : x3.5
```

The effect on the Phase 1 numbers

On the 2016-2018 screen of 465 pairs, the count of pairs called significant at the 5 % level falls from **112** with `adfuller` to **47** with `coint`. Sixty-five discoveries evaporate.

The expected-false-positive figure quoted in Phase 1 (about 23, from Proposition 7 of that document) was correct for a valid 5 % test — but the screen it was being compared against was running at an effective level of 14.6 %, where the expected count is 68. The comparison was meaningless.

Every Phase 2 function uses `coint`. The Phase 1 notebooks are left as written, as a record of what was done, with the correction flagged in the README.

6. The half-life estimator is biased, and the bias is predictable

The naive expectation is that the half-life estimator returns NaN on a random walk, since $\varphi \geq 1$ admits no solution to $\varphi^h = 1/2$. It does not. On 4000 simulated random walks of 750 observations it returned NaN only 4 % of the time, with a median estimate of 111 days. The reason is a classical result.

Theorem 6 (Small-sample bias of the AR(1) coefficient; Kendall 1954, Marriott and Pope 1954)

For the AR(1) model estimated with an intercept on T observations,

$$\mathbb{E}[\hat{\varphi}] - \varphi \approx -\frac{1+3\varphi}{T} + O(T^{-2}).$$

The estimator is biased **downward**, and the bias is largest exactly where it hurts: near the unit root.

Proposition 4 (The phantom half-life grows linearly in the sample size)

Applying Theorem 6 at $\varphi = 1$ gives $\hat{\varphi} \approx 1 - 4/T$, whence

$$h_{\text{phantom}} = -\frac{\ln 2}{\ln(1 - 4/T)} \approx \frac{\ln 2}{4} T \approx 0.173 T.$$

So a pure random walk does not merely produce a finite half-life: it produces one **proportional to the estimation window**.

Proof.

Substitute $\varphi = 1$ in Theorem 6 to get the leading bias $-4/T$. Then expand the logarithm for large T , $\ln(1 - 4/T) = -4/T + O(T^{-2})$, and substitute into the half-life formula of Proposition 2. ■

Theory and measurement can now be compared directly. The agreement is close enough to confirm the mechanism, and the residual gap has a clear cause: the sampling distribution of $\hat{\varphi}$ is skewed, so the median of $h(\hat{\varphi})$ sits below $h(\mathbb{E}[\hat{\varphi}])$ by Jensen's inequality.

Sample size T	$\hat{\varphi}$ measured	$\hat{\varphi}$ predicted ($1 - 4/T$)	h predicted	h median measured
250	0.97837	0.98400	43.0 d	37.4 d
500	0.98913	0.99200	86.3 d	74.5 d
750	0.99284	0.99467	129.6 d	115.2 d
1500	0.99649	0.99733	259.6 d	227.1 d

Kendall's formula predicts the phantom half-life to within about 12 %. 3000 simulations per row.

Two consequences

A diagnostic. Proposition 4 says the estimate should scale with T on a random walk and be stable in T on a genuine mean-reverting spread. A half-life that changes when you change the estimation window is therefore a red flag, and a cheap one to check.

A rule. The half-life must never be used to decide *whether* a spread reverts — that is the cointegration test's job. It is a tradability filter, applied afterwards. And its upper bound must be tight: 18.4 % of random walks pass a 60-day cap, against 2.7 % at 30 days.

7. The z-score: causality, and an inflation nobody warns you about

7.1 What causal means, formally

Let (F_t) be the filtration generated by the observable price history up to and including date t .

Definition 2 (Causal signal)

A signal (z_t) is *causal* if z_t is F_t -measurable for every t .

The full-sample z-score, $z_t = (s_t - \text{mean of } s) / (\text{sd of } s)$ with moments taken over the whole period, is F_T -measurable and not F_t -measurable. It is therefore not a signal at all: it encodes the spread's eventual average level, which is exactly the quantity the strategy is supposed to be betting on. This is the single most common leak in pairs trading.

Measurability is not a property one can eyeball, so it is enforced by a test. `test_rolling_zscore_is_causal` rewrites the series after some index k and asserts that every z-score up to k is bit-identical. If anyone ever substitutes a full-sample or centred window, the suite fails immediately. A companion test keeps the naive estimator as an executable counter-example.

7.2 Why a rolling window is worse than it looks

A rolling window is causal, so the obvious fix is to standardise with a rolling mean and standard deviation. That is what most published implementations do. On a strongly autocorrelated spread it is badly biased, and — contrary to the usual warning — the bias **inflates** the signal rather than damping it.

Proposition 5 (Effective sample size of a rolling mean)

Let (s_t) be stationary AR(1) with parameter ϕ and variance γ , and let m_W be the mean of W consecutive observations. Then

$$\text{Var}(m_W) = \frac{\gamma}{W^2} \left(W + 2 \sum_{k=1}^{W-1} (W-k)\phi^k \right) \equiv \frac{\gamma}{W_{\text{eff}}},$$

and as W grows large,

$$W_{\text{eff}} \longrightarrow W \frac{1-\phi}{1+\phi}.$$

Proof.

Expand the double sum and group by lag. The autocovariance at lag k is $\gamma\phi^{|k|}$, and among the W^2 ordered pairs of indices exactly W have lag 0 and $2(W-k)$ have lag $\pm k$, giving the stated expression. For the limit, note that the sum equals $W\phi/(1-\phi) + O(1)$, so

$$\text{Var}(m_W) \approx \frac{\gamma}{W} \left(1 + \frac{2\phi}{1-\phi} \right) = \frac{\gamma}{W} \cdot \frac{1+\phi}{1-\phi}.$$

■

The asymptotic form is not usable in our range

At a 30-day half-life, $\phi = 0.977$ and $1/(1-\phi) = 44$, so a 60-day window is nowhere near the regime W much greater than $1/(1-\phi)$. The asymptotic formula gives $W_{\text{eff}} = 0.69$ where the exact expression gives **1.51** — an error of a factor two, and in the direction that would have made the problem look worse than it is. The exact sum was used throughout and checked against simulation.

Half-life	ϕ	W_eff, 60-day window	W_eff, 750-day in-sample window
5 d	0.8706	4.72	52.4
10 d	0.9330	2.72	26.5
15 d	0.9548	2.09	17.8
20 d	0.9659	1.79	13.5
30 d	0.9772	1.51	9.2
45 d	0.9847	1.33	6.3

Exact effective sample sizes from Proposition 5, verified by simulation. A 60-day rolling window carries 1.5 effective observations at a 30-day half-life: the rolling mean is essentially the spread itself.

7.3 The direction of the bias, measured

With so few effective observations the rolling mean tracks the spread, so the numerator of the z-score shrinks. The usual warning stops there and concludes that the signal is damped. It is not: the rolling **standard deviation** shrinks faster, because it measures the *local* dispersion of a highly autocorrelated series rather than its unconditional dispersion. The quotient therefore exceeds one.

Half-life	numerator ratio	denominator ratio	net effect on z	share of days $ z > 2$
5 d	0.98	0.88	1.12	6.9 % (true 4.6 %)
10 d	0.94	0.77	1.21	8.8 %
20 d	0.84	0.64	1.31	10.7 %
30 d	0.76	0.56	1.36	11.7 %
45 d	0.69	0.49	1.40	12.1 %

Rolling window $W = 60$, simulated OU paths. At a 30-day half-life the strategy fires roughly 2.5 times too many entries.

7.4 The design decision

The remedy is to freeze μ and σ on the in-sample window — the same window that produced β and the cointegration p-value — and apply them unchanged through the trading year. This is exactly as causal, has no small-sample tracking problem, and keeps the entire calibration on one window.

$$z_t = \frac{s_t - \hat{\mu}_{IS}}{\hat{\sigma}_{IS}}, \quad (\hat{\mu}_{IS}, \hat{\sigma}_{IS}) \text{ estimated on the in-sample window only.}$$

This raises the effective sample size from 1.5 to 9.2 at a 30-day half-life and roughly halves the distortion: the share of days with $|z| > 2$ falls from 11.4 % to 8.0 %, against 4.6 % for the ideal. It does **not** eliminate it, because the in-sample window itself carries only about nine effective observations at that half-life.

Two things this forces us to admit

An entry threshold of $|z| > 2$ is not a reliable prior on trade frequency. Realised turnover must be measured in Phase 6, not assumed from the Gaussian tail probability.

The half-life cap does triple duty. It bounds turnover, it keeps random walks out (Section 6), and it keeps the in-sample effective sample size large enough for σ to be estimable at all. On that last count $h \leq 20$ ($W_{\text{eff}} = 13.5$) is comfortable and $h = 30$ ($W_{\text{eff}} = 9.2$) is the edge. The default cap of 30 is a documented compromise, not a comfortable margin — and a sensitivity analysis on it belongs in Phase 7.

What we give up by freezing: the z-score cannot absorb a shift in the spread's level. That is deliberate. Silently absorbing a level shift is precisely how a broken pair keeps generating signals; detecting it is the job of the regime machinery in Phase 4, not of the normalisation.

8. The quality filter

The cointegration test says a spread is stationary. It does not say the spread is tradable. Two bands, applied **after** the test and **in-sample only**, both calibrated on evidence rather than taste.

8.1 Half-life band: 2 to 30 days

- **Below 2 days:** the signal is dominated by microstructure noise and bid-ask bounce, and the implied turnover lets transaction costs consume the edge. Phase 6 quantifies this.
- **Above 30 days:** three arguments converge, and they are independent of one another. *Economic* — the out-of-sample window is one year, so a 60-day half-life allows only about four reversion cycles, too few round trips to say anything statistical and a long exposure to the relation breaking first. *Statistical* — by Proposition 4 a loose cap admits random walks: 18.4 % pass at 60 days against 2.7 % at 30. *Estimation* — by Proposition 5 the in-sample effective sample size falls to 9.2 at 30 days, which is already marginal for estimating σ .

8.2 Hedge-ratio band: 1/3 to 3

A β outside a sane band is not a hedge; it is a directional bet on one leg with the other as decoration.

- $\beta \leq 0$ means going long *both* legs. The position carries full sector exposure and market neutrality — the entire premise — is gone. On the 2016-2018 screen all five negative- β pairs involved PCG (collapsing on wildfire liabilities) or SCG (the nuclear-project scandal). A negative β fits two names trending in opposite directions and means nothing economically.
- **β near 0, or symmetrically near infinity**, means a degenerate regression. This bites when the legs have very different volatility: by Theorem 1, $\beta = S_{ab}/S_{bb}$ shrinks toward zero as the regressor's variance grows, and the residual becomes little more than a rescaled copy of the dependent variable. On the same screen **NRG** — a merchant generator, far more volatile than the regulated names — appeared in **13 of 41** significant pairs (32 %, against 3.2 expected under a uniform spread), every one with β between 0.13 and 0.30.

The band is symmetric under leg inversion, 1/3 and 3, so the filter does not inherit the arbitrariness of the alphabetical convention. A unit test enforces that symmetry.

9. Two findings that set up Phase 3

9.1 Discoveries barely exceed chance, and twice not at all

Proposition 7 of the Phase 1 document gives the expected number of spurious discoveries under the null, $E[V] = \alpha N$, with no independence assumption. Comparing the observed count against it is the cheapest available sanity check, and on this universe it is sobering.

In-sample window	Pairs tested	Significant	Expected under H0	Excess
2014-2016	351	51	18	+33
2015-2017	378	20	19	+1
2016-2018	406	41	20	+21
2017-2019	378	28	19	+9
2018-2020	378	39	19	+20
2019-2021	378	94	19	+75
2020-2022	406	95	20	+75

In-sample window	Pairs tested	Significant	Expected under H0	Excess
2021-2023	406	25	20	+5
2022-2024	465	23	23	-0

Two folds find nothing beyond pure chance. The 2019-2021 and 2020-2022 spikes sit squarely on the COVID shock, where a common shock makes everything co-move and manufactures spurious cointegration.

Two folds — 2015-2017 and 2022-2024 — find nothing beyond what a 5 % test produces on pure noise. Selecting pairs there means trading noise, and a strategy that trades them regardless will show it out of sample. Phase 7 will have to report those folds separately rather than aggregate them away.

9.2 The tests are not independent, and the selection is concentrated

In **every** fold, one name holds between 16 % and 50 % of the selected pairs — and it is a different name each time: D, SRE, AEP, AEE, ETR, AWK. This is not one odd stock. It is a property of the method: any name whose idiosyncratic path happens to look mean-reverting against the sector in-sample becomes a hub.

Fold	Pairs kept	Most frequent name	Its share
2014-2016	50	D	26 %
2015-2017	18	D	33 %
2016-2018	24	SRE	50 %
2017-2019	27	SRE	37 %
2018-2020	39	SRE	26 %
2019-2021	53	AEP	25 %
2020-2022	76	AEE	16 %
2021-2023	19	ETR	37 %
2022-2024	12	AWK	25 %

Why this is the hinge between Phase 2 and Phase 3

Risk. A selection of 24 pairs with one name in 12 of them is not 24 bets. It is one bet expressed twelve ways. Sizing that treats the pairs as independent silently concentrates risk, which Phase 6 must handle.

Statistics. Recall that $E[V] = \alpha N$ holds under arbitrary dependence, but $\text{Var}(V)$ does not. Every name shares the sector factor, and a hub induces strong positive dependence across all pairs containing it, so the effective number of independent tests is far below $N(N-1)/2$.

That is exactly the regime where the Phase 3 choice stops being cosmetic: **Bonferroni** controls the family-wise error rate under *any* dependence structure, by the union bound; **Benjamini-Hochberg** controls the false discovery rate under independence or positive regression dependence, but not in general. A permutation or block-bootstrap null, which reproduces the dependence rather than assuming it away, is the honest third option.

10. What Phase 2 deliberately does not do

- **No selection.** `screen_pairs` runs the tests; it does not decide what to trade. Selection, the multiple-testing correction and the walk-forward loop are Phase 3.
- **No time awareness.** Nothing in `src/cointegration.py` knows about dates, so nothing there prevents it being handed out-of-sample data. That discipline lives in the walk-forward loop, not in the estimator — a module that cannot see dates cannot silently cheat with them.
- **No liquidity filter.** Still deferred, for the Phase 1 reason: it must be recomputed fold by fold on the in-sample window only, or it becomes survivorship bias dressed as a technical criterion.

Interview questions this phase should let you answer

- Derive the half-life from an AR(1). What is its continuous-time counterpart, and why are they the same number?
- Why can you not test cointegration with an ordinary t-statistic? What goes wrong, and at what rate?
- You ran ADF on an OLS residual. What is the size of that test, and why is it not 5 %?
- Your half-life estimator returns 111 days on a random walk. Explain, and give the scaling in T.
- Why is a rolling z-score biased on a mean-reverting spread, and in which direction?
- Your screener finds 47 significant pairs and expects 23 by chance. What do you conclude, and what would you do next?
- Bonferroni is valid under arbitrary dependence and BH is not. Why is that not an argument for using Bonferroni here?

11. Notation

Symbol	Meaning
a_t, b_t	Total-return log-prices of the two legs
s_t	Spread, $a_t - \beta b_t - \alpha$
β	Hedge ratio; β 's estimate
φ	AR(1) autoregressive parameter; $\varphi = 1$ is a unit root
θ	OU mean-reversion speed; $\varphi = \exp(-\theta\Delta)$
h	Half-life, $h = -\ln 2 / \ln \varphi = \ln 2 / \theta$
γ	Unconditional variance of the spread
W_{eff}	Effective sample size of a rolling mean (Proposition 5)
T	Number of in-sample observations
N	Number of pairs tested, $N = n(n-1)/2$
V	Number of false discoveries
$W(u)$	Standard Brownian motion on $[0,1]$
F_t	Information available at date t